

Design of Linear Amplifiers using BJTs

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Linear amplifiers are analogue circuits. Proper operation of any analogue circuit requires that DC components (voltages or currents) are added to the voltages and currents inside this circuit. The DC components exist independently of any signal fluctuations and do not constitute signal information passing through the circuit.

The term signal is thus used to denote only the information-carrying fluctuations of a given voltage or current. Any fixed DC levels upon which such signals are superimposed are called *bias* components. The design and analysis of an analogue circuit generally requires that the total value of a voltage or current (signal + bias) be considered, even though only the signal component may be of interest.

Biasing is most often used in circuits that contain non-linear devices like diodes and transistors. When properly implemented, biasing causes these non-linear elements to behave as linear elements, thus greatly enhancing their usefulness.

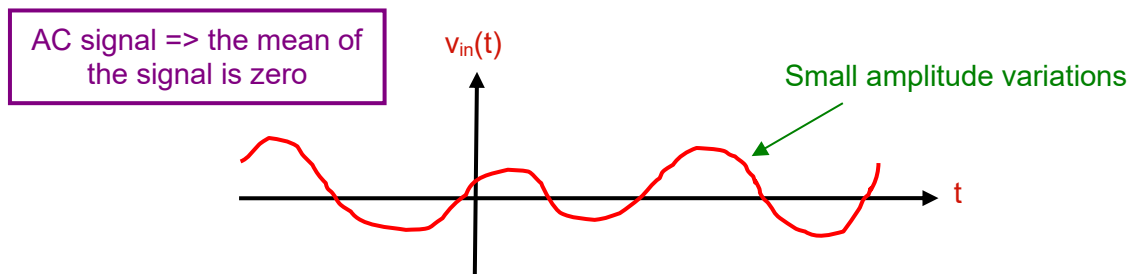
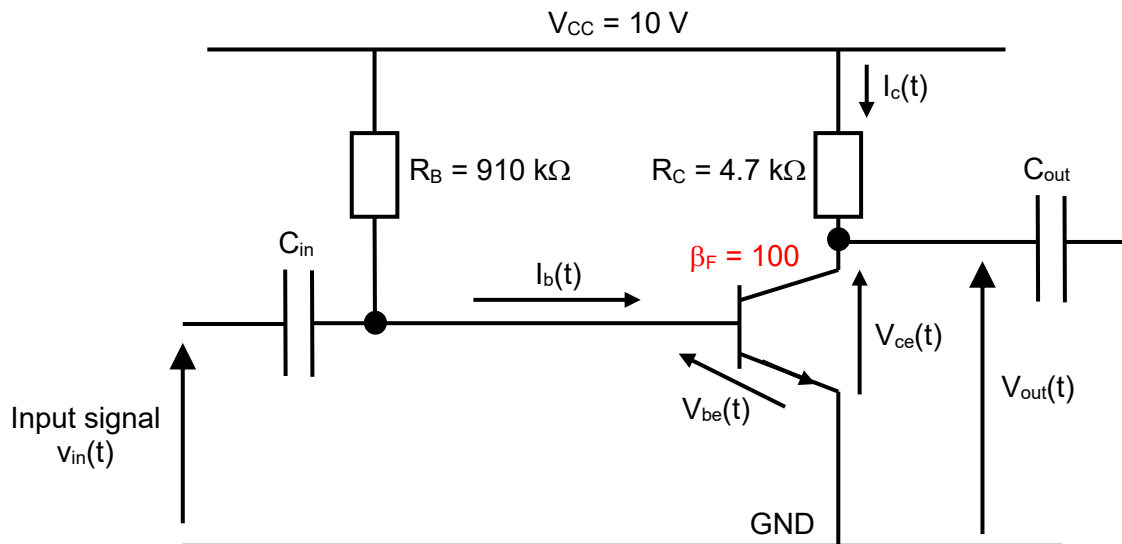
Although the various characteristics of most semi-conductor devices are non-linear, many exhibit linear behaviour over certain regions of operation. For instance, the transfer characteristic of a BJT can be described by the linear equation $I_C = \beta_F I_B$ in the forward active of operation.

The technique of biasing is employed by the circuit designer to confine a device's operating point to a region where its behaviour is linear (or at least approximately linear) while avoiding the gross non-linearities in the device's characteristics.

Throughout this Chapter, we consider the example of a common-emitter (CE) amplifier in order to show how to design and study linear amplifiers using bipolar junction transistors (BJTs).

• Study of a Common-Emitter (CE) Amplifier

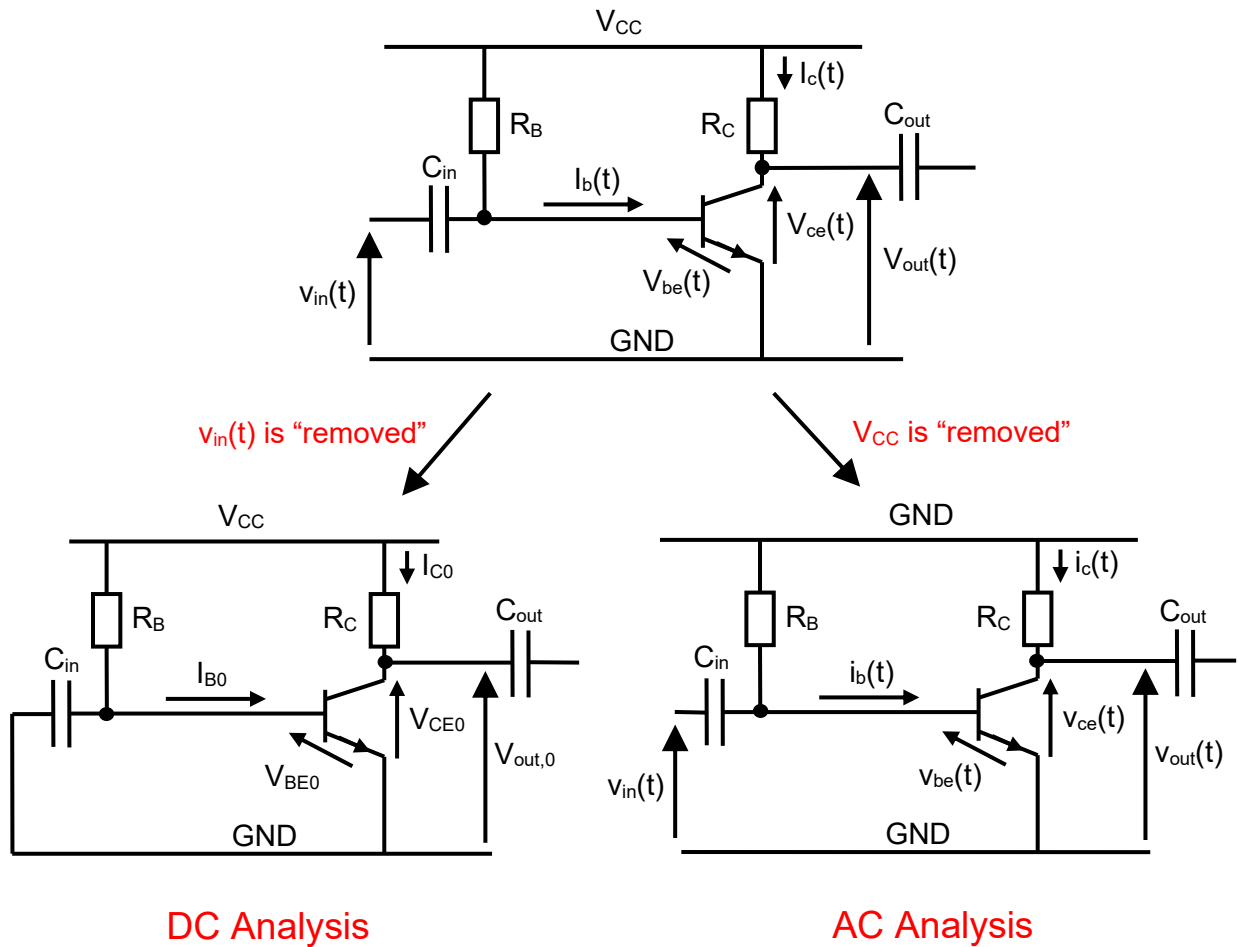
Consider the CE amplifier below. As its name implies, this circuit can be used to amplify an input signal $v_{in}(t)$ that is represented by the (small) variation of an input voltage with time. For the time being, we assume that the mean of $v_{in}(t)$ is zero, i.e. $v_{in}(t)$ is an AC signal.



Note that we use two capacitors C_{in} and C_{out} , called *coupling capacitors*, in this circuit.

Apart from the AC signal $v_{in}(t)$ to be amplified, the CE amplifier has another input: the supply voltage V_{CC} . It is the DC (constant) voltage source that provides the amplifier with the power necessary to boost the input voltage. It also allows the BJT to operate in the forward active mode of operation.

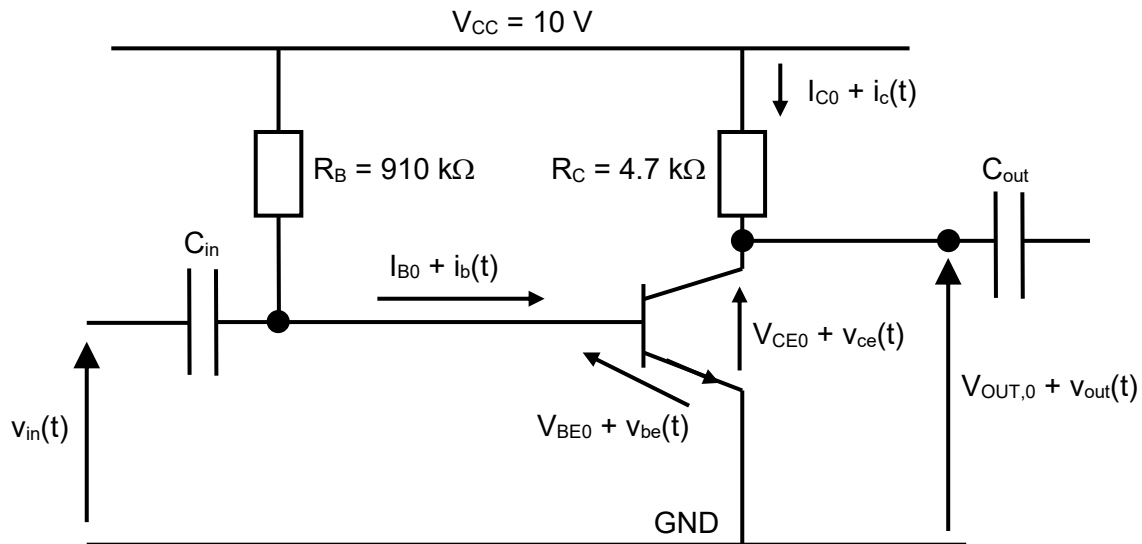
Since there are two inputs, we can use the *principle of superposition* to determine the expressions of the various currents and voltages in the circuit.



According to the principle of superposition, any signal $X(t)$ in the circuit is therefore the sum of a DC component X_0 (mean of $X(t)$) and an AC component $x(t)$, i.e. $X(t) = X_0 + x(t)$.

The quantity X_0 is determined by the DC analysis of the circuit, whereas the expression of $x(t)$ is determined by the AC analysis of the circuit. For instance, we can write:

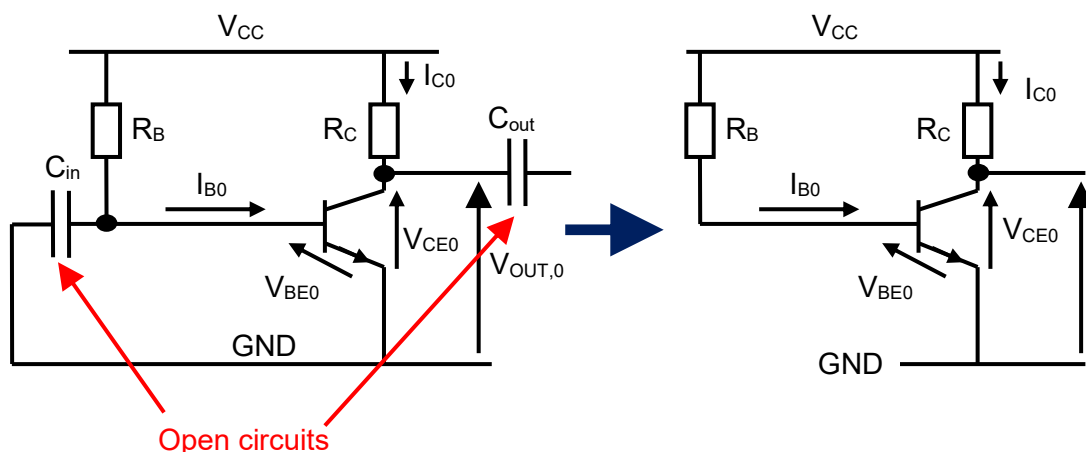
- $I_b(t) = I_{B0} + i_b(t)$ and $I_c(t) = I_{C0} + i_c(t)$;
- $V_{be}(t) = V_{BE0} + v_{be}(t)$ and $V_{ce}(t) = V_{CE0} + v_{ce}(t)$;
- $V_{out}(t) = V_{OUT,0} + v_{out}(t)$.



• DC analysis of our CE Amplifier

We are now going to perform the DC analysis of our CE amplifier. The main purpose of the DC analysis is to determine whether or not the BJT is properly biased.

Bipolar transistors have four different modes of operation. A BJT must be and always remain in the forward active mode when used to design an amplifier, whatever the value of the input signal to be amplified. Biasing an amplifier circuit consists of connecting the BJT so that it operates in the forward active mode.

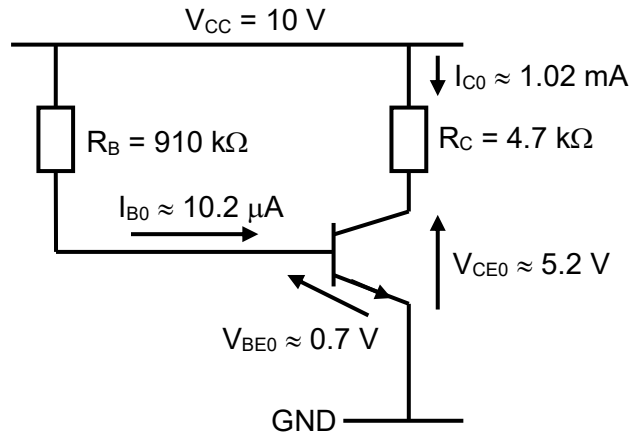


The base-emitter junction is on. We have $V_{BE0} = V_{BE,on} \approx 0.7$ volt. Therefore, the base current is

$$\text{given by } I_{B0} = \frac{V_{CC} - V_{BE,on}}{R_B} \approx 10.2 \mu\text{A}.$$

Assume now that the BJT is in the forward active mode (we actually do not know it for the time being, we will check later whether we were right or not). We also assume that the forward current gain of the BJT is given by $\beta_F = 100$.

We can thus write $I_{C0} = \beta_F I_{B0} \approx 1.02 \text{ mA}$. Then, we can compute the value of the voltage V_{CE0} as $V_{CE0} = V_{CC} - R_C I_{C0} \approx 5.2 \text{ volts}$.



Since $V_{CE0} > V_{CE,sat}$, we conclude that our assumption was right and the BJT is indeed in the forward active mode.

In addition, we can show that the BJT is in fact ideally biased as it operates midway between the “start of the saturation mode” and the “start of the cut-off mode”. This maximises the voltage swing for the output voltage.

The saturation mode starts when $V_{ce}(t) = V_{CE,sat} \approx 0.2 \text{ volt}$, i.e. $V_{out}(t) \approx 0.2 \text{ volt}$.

The cut-off mode starts when $i_c(t) = 0$, i.e. $V_{ce}(t) = V_{out}(t) = V_{CC} - R_C I_c(t) = V_{CC} = 10 \text{ volts}$. Those results mean that the maximum output voltage swing is equal to $V_{CC} - V_{CE,sat} \approx 9.8 \text{ volts}$.

To achieve this maximum output voltage swing, we need to have

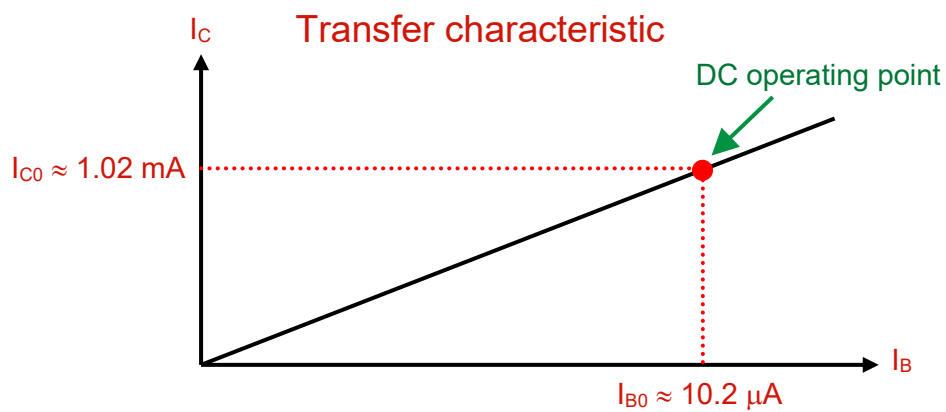
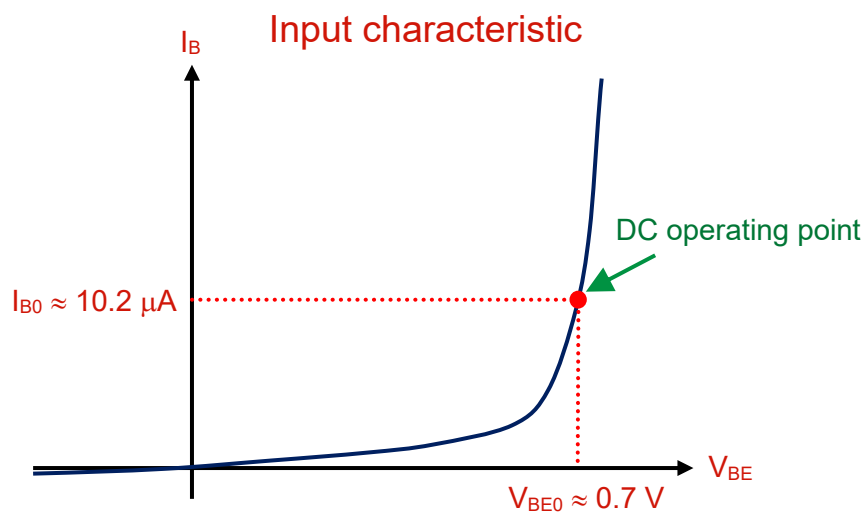
$$V_{CE0} = V_{OUT,0} = \frac{V_{CC} + V_{CE,sat}}{2} \approx 5.1 \text{ volts},$$

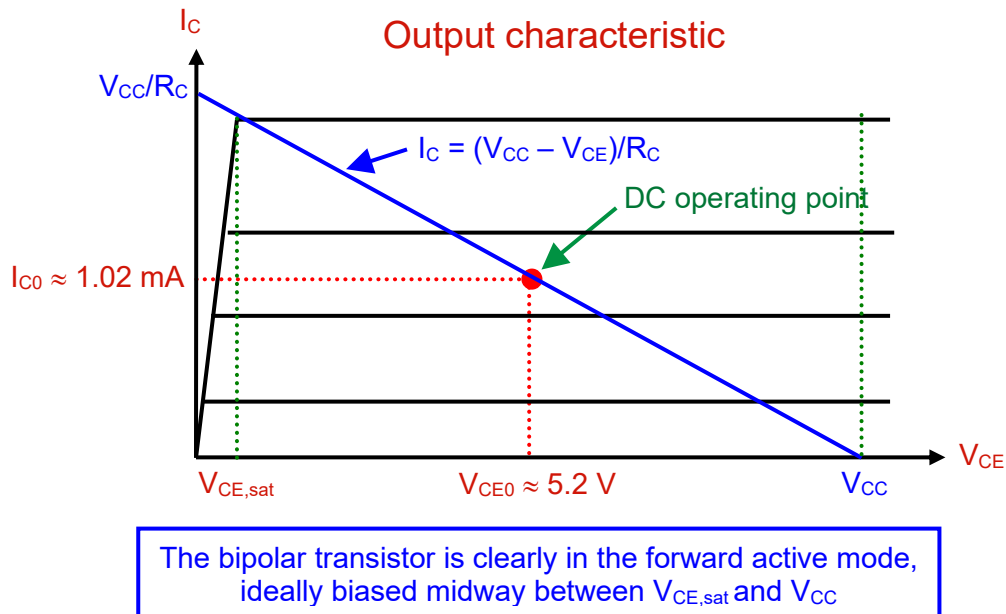
which is the midpoint between $V_{CE,sat} \approx 0.2 \text{ volt}$ (saturation) and $V_{CC} = 10 \text{ volts}$ (cut-off). This is ideal since it allows the AC signal $v_{ce}(t) = v_{out}(t)$ to swing by nearly $V_{CC}/2 = 5 \text{ volts}$ in either direction.

Since we found $V_{CE0} \approx 5.2$ volts for our CE amplifier, we can conclude that our BJT is (almost) perfectly biased.

It is very useful (and easy) to remember that, for a CE amplifier relying on a single voltage supply V_{CC} , the optimal bias corresponds to

$$V_{CE0} = \frac{V_{CC} + V_{CE,sat}}{2} \approx \frac{V_{CC}}{2}.$$

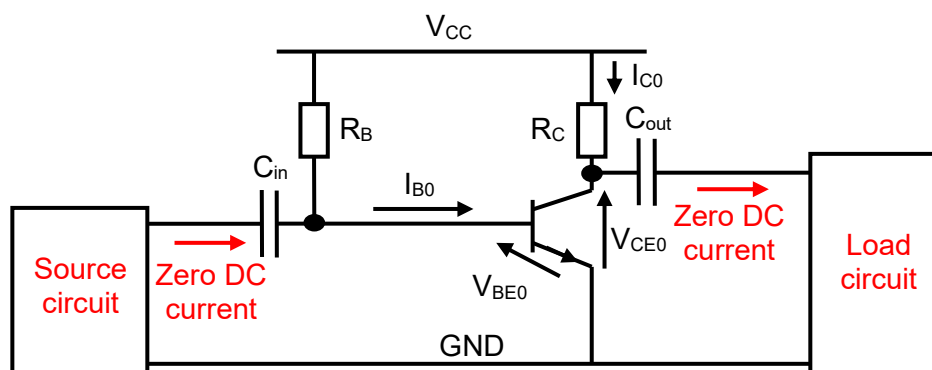




• Role Played by the Coupling Capacitors in DC Operation

The presence of both capacitances C_{in} and C_{out} protects the biasing circuit from any external influence. In other words, any change in the external environment of the amplifier circuit does not have any effect on the DC operating points. This is simply due to the fact that no DC current can flow through a capacitance.

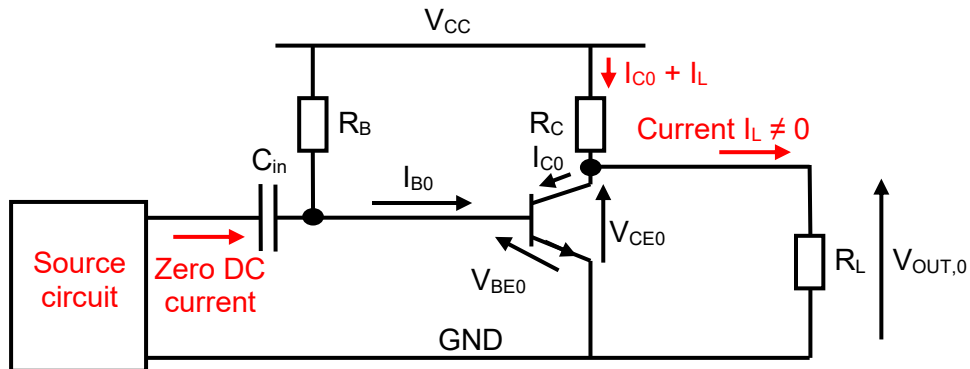
To clarify this point, let us assume that our CE amplifier circuit is connected to a source circuit and a load circuit, as shown below. Note that this will always be the case in practice.



It clearly appears that, thanks to the presence of both coupling capacitors, the DC currents I_{B0} and I_{C0} as well as the DC voltages V_{BE0} and V_{CE0} are independent of the parameters of the

source and load circuits. This means that the DC operating points of our amplifier are not affected by those parameters.

If no coupling capacitors were used, the DC currents I_{B0} and I_{C0} as well as the DC voltages V_{BE0} and V_{CE0} would depend on the parameters of the source and load circuits, which is unacceptable in many cases. For instance, let us investigate what would happen if no capacitance C_{out} was employed. In addition, let us represent the loading circuit by its input resistance R_L .



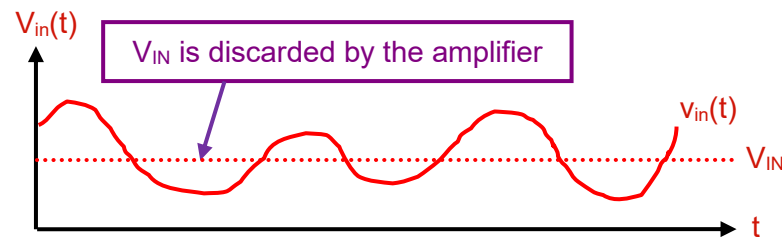
The DC operating points would be determined using the following set of equations:

- $V_{BE0} = V_{BE,on} \approx 0.7$ volt.
- $I_{B0} = \frac{V_{CC} - V_{BE,on}}{R_B} \approx 10.2 \mu A$.
- $I_{C0} = \beta_F I_{B0} \approx 1.02$ mA.
- $V_{CE0} = V_{CC} - R_C(I_{C0} + I_L) = V_{CC} - R_C\left(I_{C0} + \frac{V_{CE0}}{R_L}\right) \Rightarrow V_{CE0} = \frac{R_L}{R_L + R_C}(V_{CC} - R_C I_{C0}) < 5.2$ volts.

The value of V_{CE0} would thus depend on the value of R_L . This implies that the location of one of the DC operating points would be affected by a change in the input resistance of the loading circuit. Depending on the value of R_L , the voltage V_{CE0} could vary from 5.2 volts (for $R_L \gg R_C$) to zero (for $R_L \ll R_C$).

It is worth mentioning that there is a price to pay for the use of coupling capacitors. The presence of both coupling capacitances C_{in} and C_{out} puts a lower limit on the frequency of the signals that can be amplified. This implies that DC signals, which are zero-frequency signals, cannot be amplified.

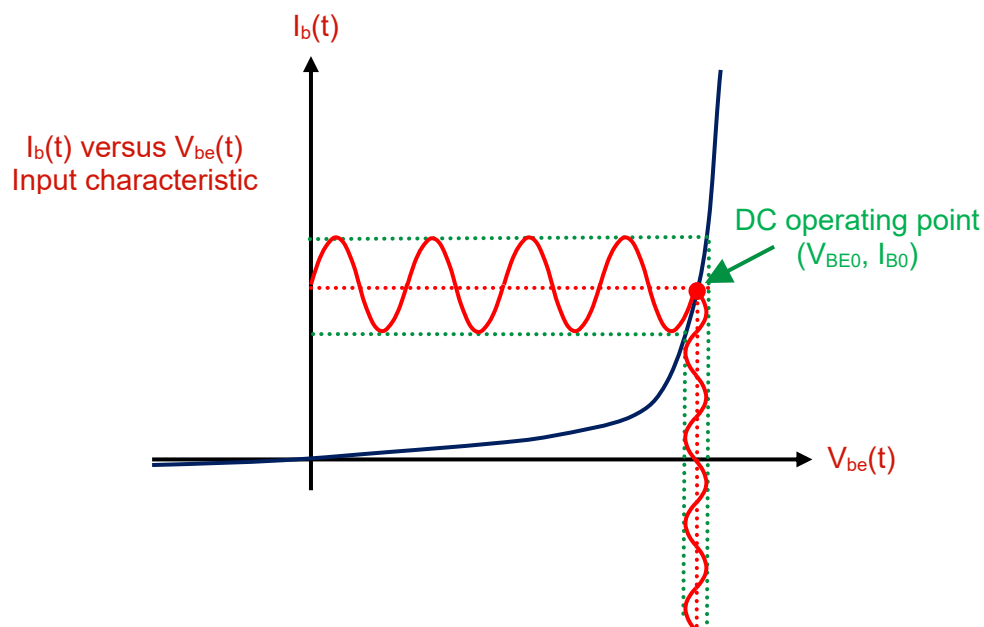
If there was a DC component in the input signal, it would be filtered out by C_{in} and ignored by the amplifier circuit as a result of this.

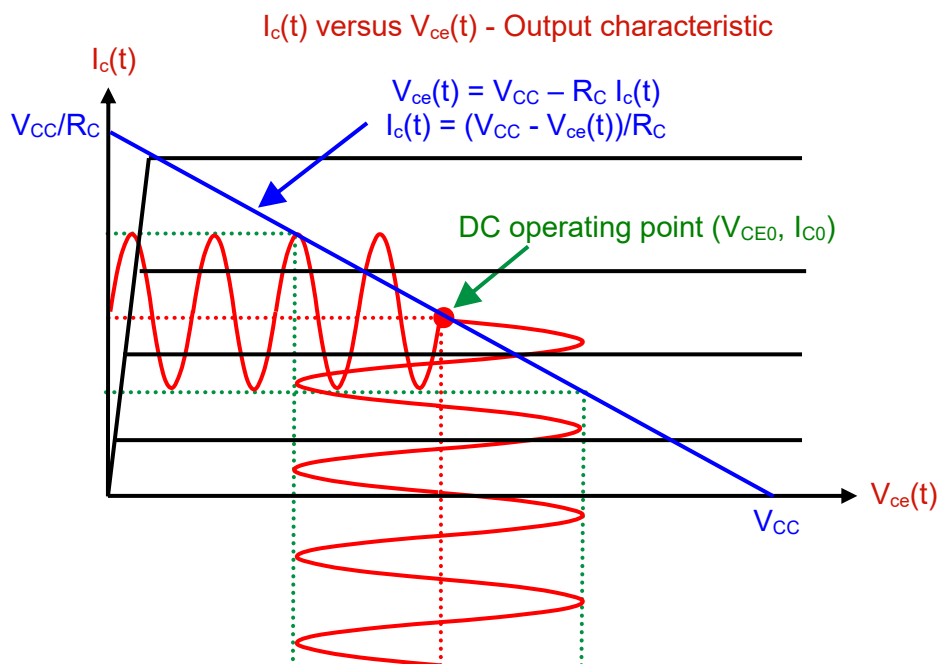
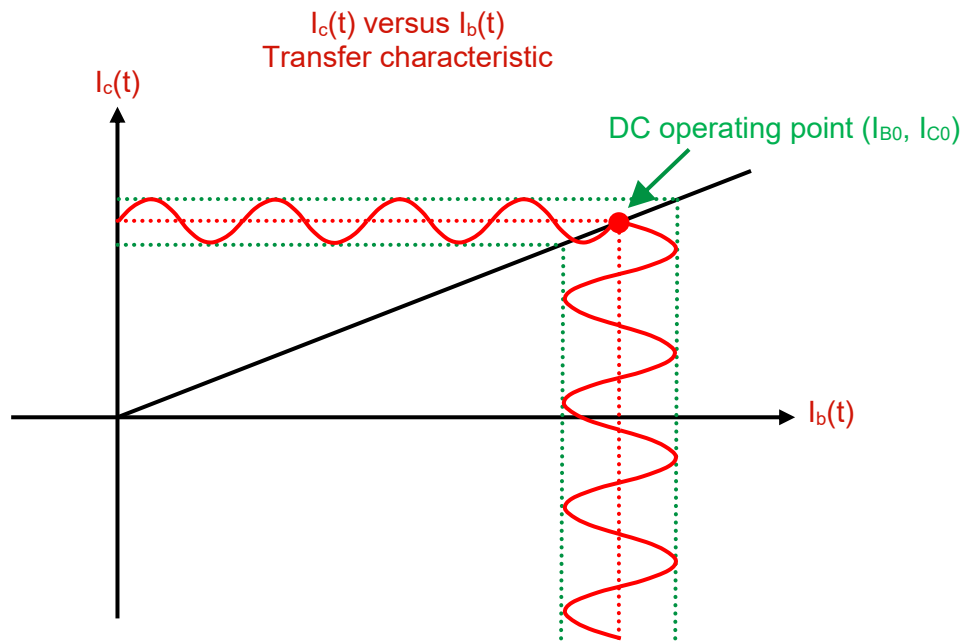


• AC analysis of our CE Amplifier

Our bipolar transistor is properly biased and can therefore be used to amplify an input voltage $v_{in}(t)$. We assume that $v_{in}(t)$ is an AC signal with small amplitude variations. As we have just seen, if the input signal also contained a DC component, the latter would be filtered out by the coupling capacitance C_{in} and would therefore be ignored by the amplifier circuit.

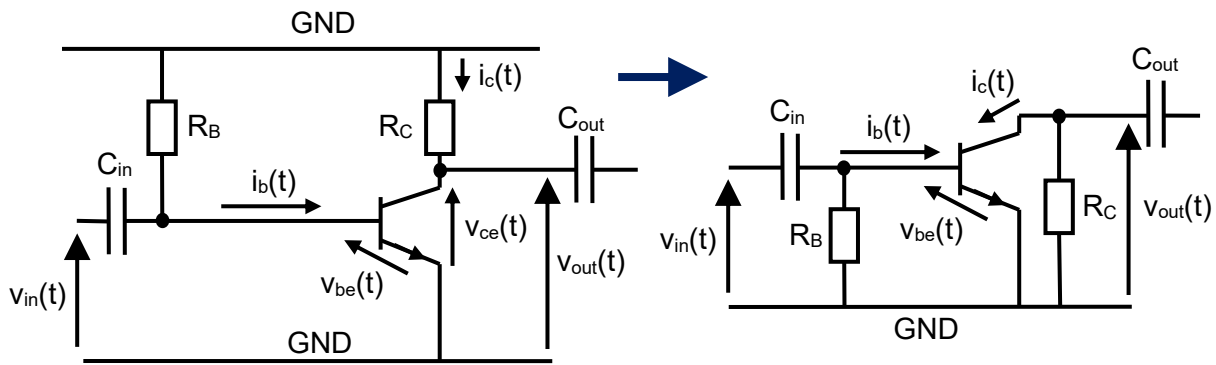
Any fluctuations of the AC signal $v_{in}(t)$ will induce corresponding AC variations of the various voltages and currents inside the circuit around their DC values, as illustrated by the three plots shown below.





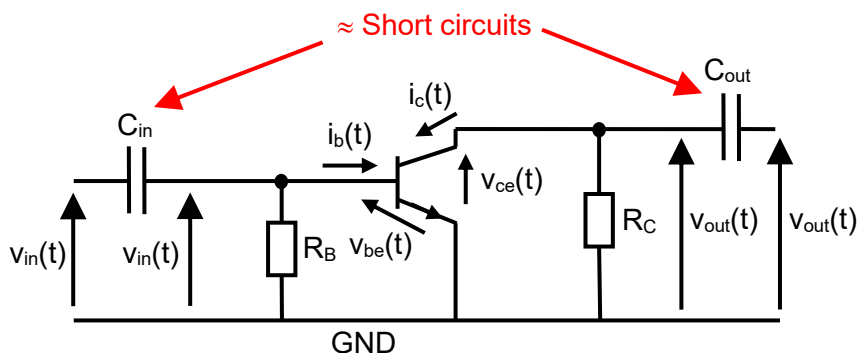
The magnitude of $v_{in}(t)$ must be sufficiently small so that the variation of $V_{ce}(t)$ around V_{CE0} allows the transistor to remain in the forward active mode of operation.

For the AC analysis of our CE amplifier, we need to study the circuit depicted below.



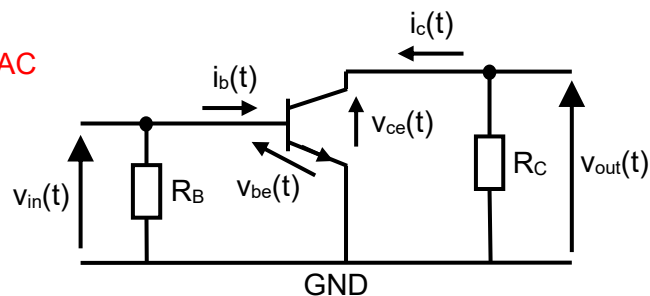
It is possible to further simplify the circuit by getting rid of the coupling capacitances C_{in} and C_{out} . To do so, we can assume that each coupling capacitance has a sufficiently high value to be considered as a short circuit (perfect wire) at the frequency of the input signal.

In other words, we can assume that both AC input and output signals $v_{in}(t)$ and $v_{out}(t)$ are passed through the capacitances without being affected at all.



Impedance of a capacitance C at a given frequency ω :
 $Z = 1/jC\omega$
 If $C\omega$ is "large enough", $Z \approx 0$

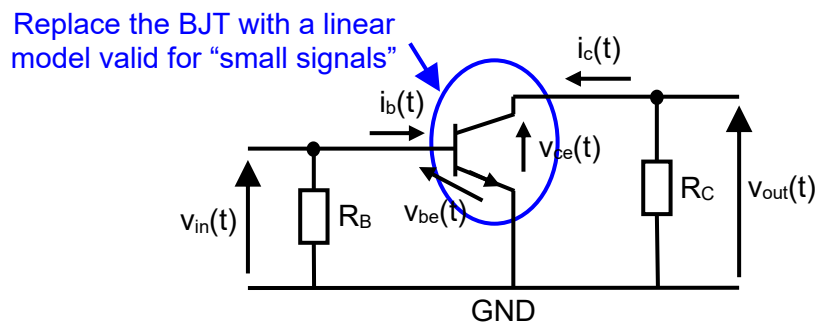
We need to perform the AC analysis of this circuit



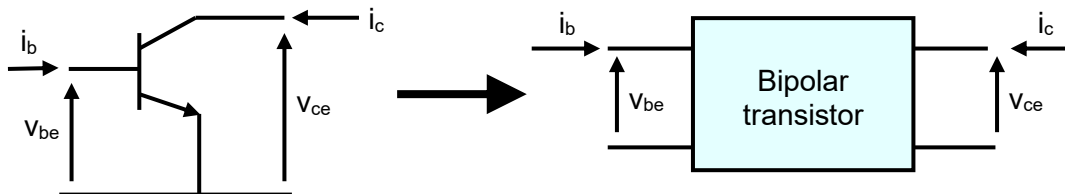
The unloaded voltage gain of our CE amplifier is defined as $A_v = \frac{v_{out}(t)}{v_{in}(t)}$.

• Small-Signal Model of a BJT

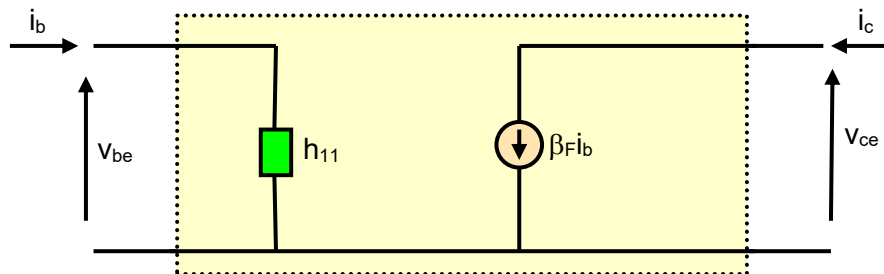
In order to perform the AC analysis of the amplifier, we need to replace the BJT with an equivalent model valid for a small-amplitude input signal (“small signal”). Obviously, this model is derived assuming that the BJT is in the forward active mode of operation.



The bipolar transistor is now seen as a “black box” with two inputs and two outputs.



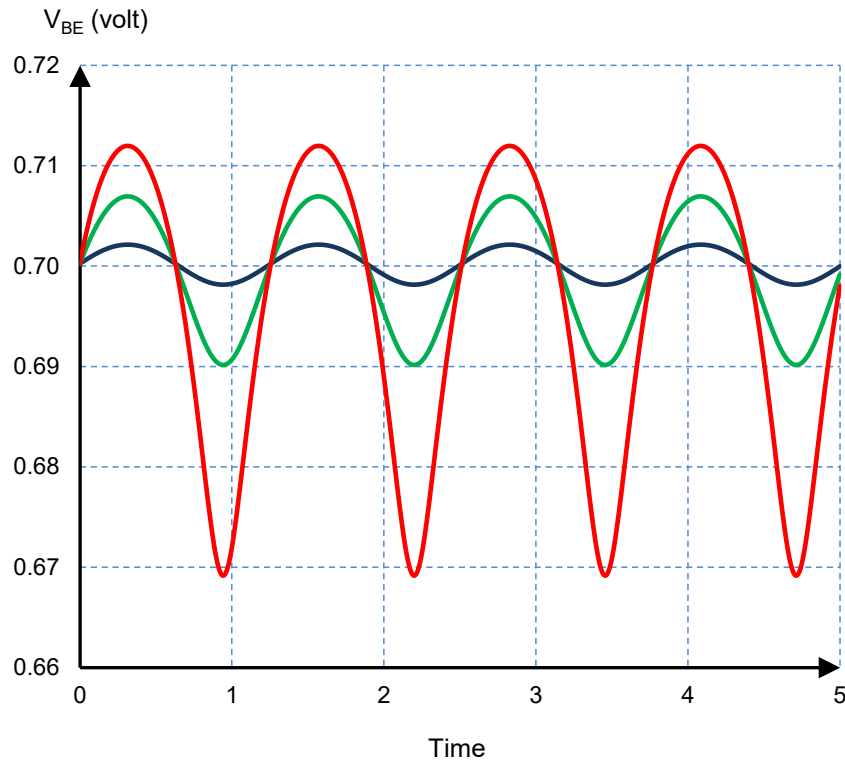
We can show that the small-signal model of a BJT is given below.



The presence of a resistance h_{11} between base and emitter implies that the relation between v_{be} and i_b is linear, i.e. $v_{be} = h_{11}i_b$, which is not the case. Remember that the base current is in fact an exponential function of the base-emitter voltage as if the base current was flowing through the base-emitter junction.

However, we must keep in mind the fact that all quantities v_{be} , v_{ce} , i_b , and i_c represent AC signals, i.e. fluctuations of the corresponding voltages and currents around the DC operating points. The assumption that these fluctuations are *small* allows us to approximate the expressions linking those quantities using linear functions. After all, any mathematical function can be replaced by a straight line over a sufficiently small interval.

To illustrate this point, let us consider the case where the base current $I_b(t) = I_{B0} + i_b(t)$ is a sinusoidal signal with a mean $I_{B0} = 10.2 \mu\text{A}$ and an amplitude A . In other words, the current $I_b(t)$ varies in a sinusoidal fashion between $I_{B0} - A$ and $I_{B0} + A$. The figure depicted below shows the corresponding variation of the base-emitter voltage $V_{be}(t) = V_{BE0} + v_{be}(t)$, for $A = 1 \mu\text{A}$ (blue curve), $A = 4 \mu\text{A}$ (green curve), and $A = 8 \mu\text{A}$ (red curve).

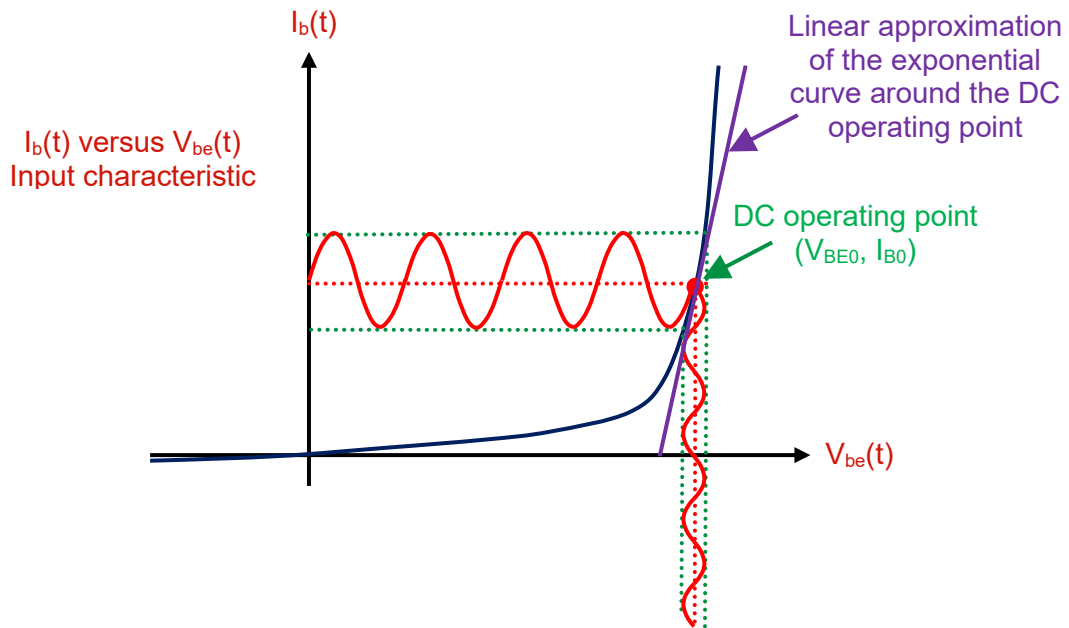


These results indicate that, as the amplitude A of the base current increases, so does the amplitude of $V_{be}(t)$ (as was expected). We also notice that, as A is increased, the corresponding variation of $V_{be}(t)$ looks less and less like a sinusoidal signal, thus implying that the linearity of the relationship between $I_b(t)$ and $V_{be}(t)$ is progressively lost.

However, as long as the amplitude of the base current remains sufficiently small, the relationship between $I_b(t)$ and $V_{be}(t)$ can be considered as linear, and the exponential function that links both quantities can then be replaced, around the DC operating point, by a linear function in the form

$$V_{be}(t) = h_{11}I_b(t) + b,$$

where h_{11} and b are two constants.



Around the DC operating point, we can thus write $V_{BE0} + v_{be}(t) = h_{11}I_{B0} + h_{11}i_b(t) + b$. If we focus on the AC component, we see that $v_{be}(t) = h_{11}i_b(t)$. In electronic engineering, this equation is simply implemented using a resistance h_{11} .

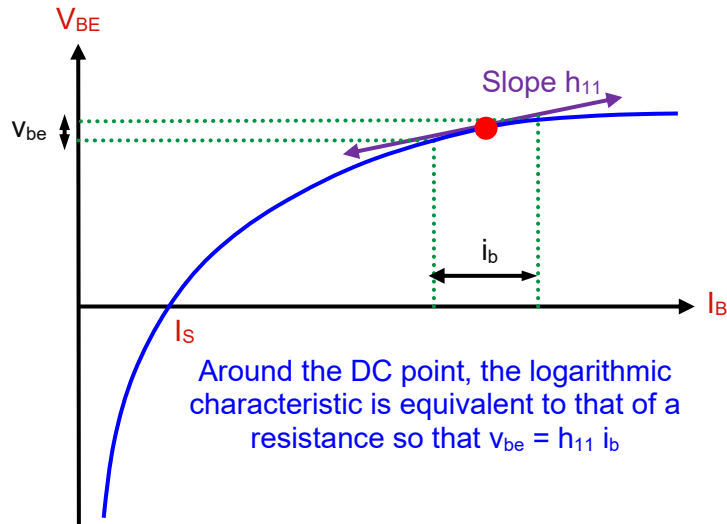
The expression of the resistance h_{11} is obtained as follows:

$$h_{11} = \frac{v_{be}}{i_b} = \left. \frac{dV_{BE}}{dI_B} \right|_{V_{BE0}, I_{B0}}.$$

The equation of the base-emitter junction characteristic is given by

$$I_B \approx I_S \cdot \exp\left\{\frac{V_{BE}}{V_T}\right\}.$$

This expression is equivalent to $V_{BE} \approx V_T \cdot \ln\left(\frac{I_B}{I_S}\right) = V_T \cdot \ln(I_B) - V_T \cdot \ln(I_S)$.



Therefore, we can write $h_{11} = \left. \frac{dV_{BE}}{dI_B} \right|_{V_{BE0}, I_{B0}} \approx \left. \frac{V_T}{I_B} \right|_{V_{BE0}, I_{B0}} = \frac{V_T}{I_{B0}}$.

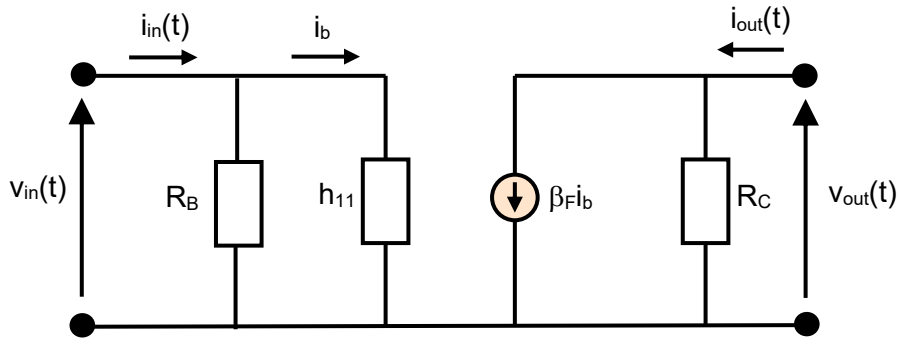
At room temperature, the expression of the parameter h_{11} is thus given by $h_{11} \approx \frac{25 \text{ mV}}{I_{B0}}$.

As for the CE amplifier considered throughout this chapter, we have $h_{11} \approx 2.45 \text{ k}\Omega$ (at room temperature) since $I_{B0} \approx 10.2 \text{ }\mu\text{A}$.

• Small-Signal AC Analysis of our CE Amplifier

We can now replace the bipolar transistor with its small-signal model in order to determine the expressions of the OC voltage gain, input resistance, and output resistance of our CE amplifier.

We recall the values of the circuit parameters: $R_B = 910 \text{ k}\Omega$, $h_{11} \approx 2.45 \text{ k}\Omega$, $\beta_F = 100$, $R_C = 4.7 \text{ k}\Omega$.



The general equations for this circuit are as follows:

(1) $v_{in}(t) = R_B (i_{in}(t) - i_b)$, (2) $v_{in}(t) = h_{11} i_b$, and (3) $v_{out}(t) = R_C (i_{out}(t) - \beta_F i_b)$.

- **OC Voltage Gain:** The OC voltage gain of our CE amplifier is given by

$$A_v = \left. \frac{v_{out}(t)}{v_{in}(t)} \right|_{i_{out}(t)=0} = -\beta_F \frac{R_C}{h_{11}} \approx -192.$$

- **Input Resistance:** The input resistance is given by $R_{in} = \frac{v_{in}(t)}{i_{in}(t)} = \frac{R_B h_{11}}{R_B + h_{11}} \approx h_{11}$ since $h_{11} \ll R_B$.

Its value is thus $R_{in} \approx 2.45 \text{ k}\Omega$.

- **Output Resistance:** The output resistance is given by $R_{out} = \left. \frac{v_{out}(t)}{i_{out}(t)} \right|_{v_{in}(t)=0} = R_C = 4.7 \text{ k}\Omega$.

We conclude that the CE amplifier seems to be a good voltage amplifier since it has a high OC voltage gain, although its input resistance might be a bit too low and its output resistance might be a bit too large for a voltage amplifier.

We mention in passing that the SC current gain of our CE amplifier is given by

$$A_i = \left. \frac{i_{out}(t)}{i_{in}(t)} \right|_{v_{out}(t)=0} = \beta_F \frac{R_B}{h_{11} + R_B} \approx \beta_F = 100.$$

This high SC current gain indicates that this CE amplifier can also be considered as a good current amplifier, although its input resistance might be a bit too large and its output resistance might be a bit too low for a current amplifier.

• Trade-Off Issues

It is worth noticing that the OC voltage gain of our CE amplifier can also be written as

$$A_v = -\beta_F \frac{R_C}{h_{11}} \approx -\beta_F \frac{R_C I_{B0}}{V_T} = -\frac{R_C I_{C0}}{V_T}.$$

This result indicates that the OC voltage gain is proportional to both the DC collector current I_{C0} and the collector resistance R_C . Hence, higher voltage gains could be achieved by increasing I_{C0} and/or R_C (within reasonable limits for obvious DC biasing reasons).

However, increasing I_{C0} would decrease the input resistance whose expression is given by

$R_{in} \approx h_{11} \approx \frac{V_T}{I_{B0}} = \frac{\beta_F V_T}{I_{C0}}$, and reducing the input resistance is not a smart thing to do with a voltage amplifier.

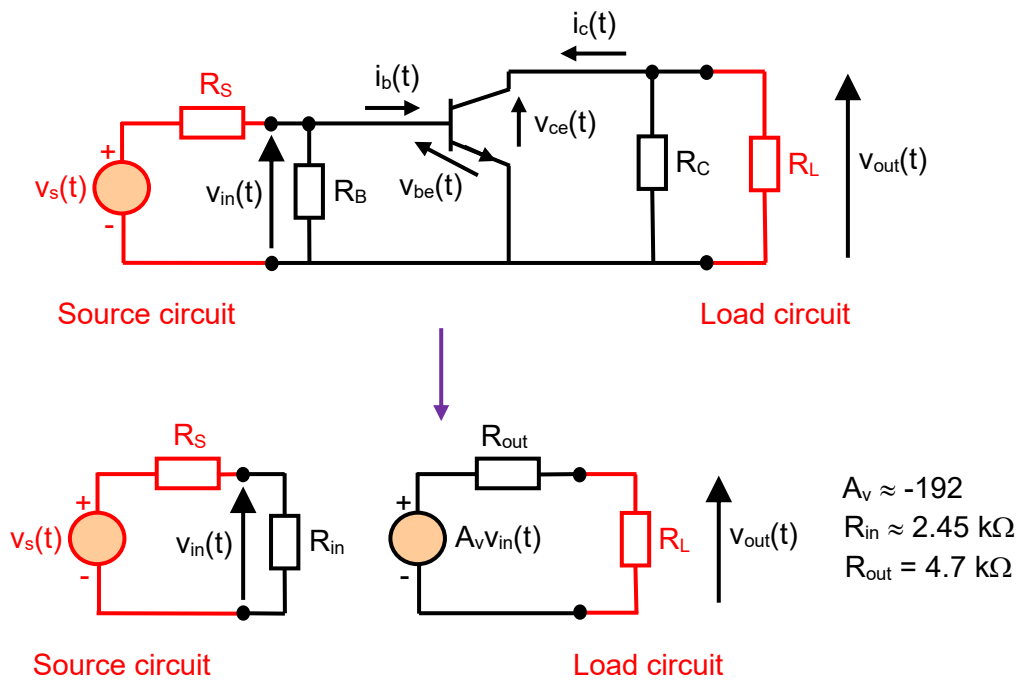
In addition, employing a higher collector resistance R_C would increase the output resistance as $R_{out} = R_C$, which is once again something to be avoided with voltage amplifiers.

Finally, it is worthwhile mentioning that any increase in the DC current I_{C0} results in an increase in the DC power consumed by the circuit (although power consumption issues are beyond the scope of this document). This could be a very big problem when the power is supplied by a battery as is the case with mobile telecommunication devices.

As a summary, there are always plenty of trade-off issues to consider when designing any electronic circuit. There is no free lunch.

• Real Voltage Gain

We have determined the value of the OC voltage gain A_v of our CE amplifier. But, what is the actual voltage gain $\tilde{A}_v$ obtained when the amplifier is connected to other circuits?



Assume that the input of our CE amplifier is connected to a voltage source $v_s(t)$ with an output resistance R_s , whereas its output is connected to a loading circuit that is represented by an input resistance R_L .

We can show that the real voltage gain of the whole circuit is given by

$$\tilde{A}_v = \frac{v_{out}(t)}{v_s(t)} = A_v \frac{R_{in}}{R_{in} + R_s} \frac{R_L}{R_L + R_{out}}.$$

Note that, if $R_{in} \ll R_s$ and/or $R_L \ll R_{out}$, we have $\tilde{A}_v \ll A_v$.

This result confirms that, in order to avoid a drastic reduction in voltage gain when the amplifier is connected to other circuits, one should maximise its input resistance R_{in} and minimise its output resistance R_{out} .

To illustrate these results, let us consider a few examples.

Assume that $R_s = R_L = 5 \text{ k}\Omega$. In this case, the value of the real voltage gain is

$$\tilde{A}_v \approx -192 \times \frac{2.45}{2.45 + 5} \times \frac{5}{5 + 4.7} \approx -192 \times 0.329 \times 0.515 \approx -32,$$

which is still a significant voltage gain.

Assuming now that $R_S = 5 \text{ k}\Omega$ and $R_L = 10 \text{ }\Omega$, the real voltage gain is given by

$$\tilde{A}_v \approx -192 \times \frac{2.45}{2.45 + 5} \times \frac{0.01}{0.01 + 4.7} \approx -192 \times 0.329 \times 0.002 \approx -0.13.$$

This very low gain is unacceptable and clearly indicates that the voltage amplifier does not work at all in that case. The collapse in voltage gain is caused by the very significant mismatch between the amplifier output resistance $R_{out} = 4.7 \text{ k}\Omega$ and the load resistance $R_L = 10 \text{ }\Omega$.

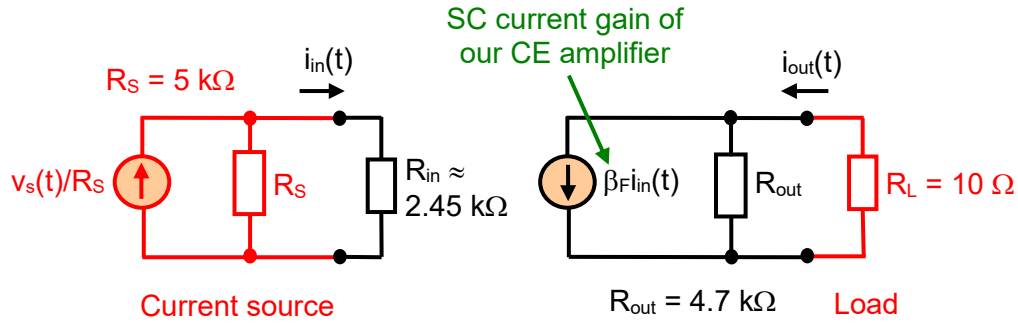
In fact, the large output resistance R_{out} (compared to R_L) means that the amplifier output port cannot provide the load circuit with a current that is sufficient to maintain a significant voltage across the load input port.

One way to solve this problem consists of inserting a specific circuit between the CE amplifier and the load. Its task is to perform an *impedance matching* function. Its input impedance must be large (ideally, $\gg 4.7 \text{ k}\Omega$) so as to minimise the loading effect on the CE amplifier, and its output impedance must be low (ideally, a few ohms at most) to minimise the loading effect due to the very low load resistance value. This circuit should actually be a good current amplifier.

A *common-collector* circuit (also known as *emitter follower*) or, even better, a Darlington-pair circuit would be suitable for this task.

The fact that no voltage amplification is achieved when $R_{in} \ll R_S$ and/or $R_L \ll R_{out}$ does not necessarily mean that the CE amplifier is useless in those cases since the circuit could then be used as a current amplifier instead.

For instance, let us consider the current amplification issue for a moment. The corresponding equivalent circuit of our CE amplifier connected to a source with $R_S = 5 \text{ k}\Omega$ and a load with $R_L = 10 \text{ }\Omega$ is shown below.



We can show that the real current gain of the whole circuit is given by

$$\tilde{A}_i = \frac{i_{out}(t)}{i_s(t)} = A_i \frac{R_S}{R_{in} + R_S} \frac{R_{out}}{R_L + R_{out}}.$$

With $R_S = 5 \text{ k}\Omega$, $R_L = 10 \text{ }\Omega$, $R_{in} \approx 2.45 \text{ k}\Omega$, $R_{out} = 4.7 \text{ k}\Omega$, and $A_i = \beta_F = 100$, we obtain

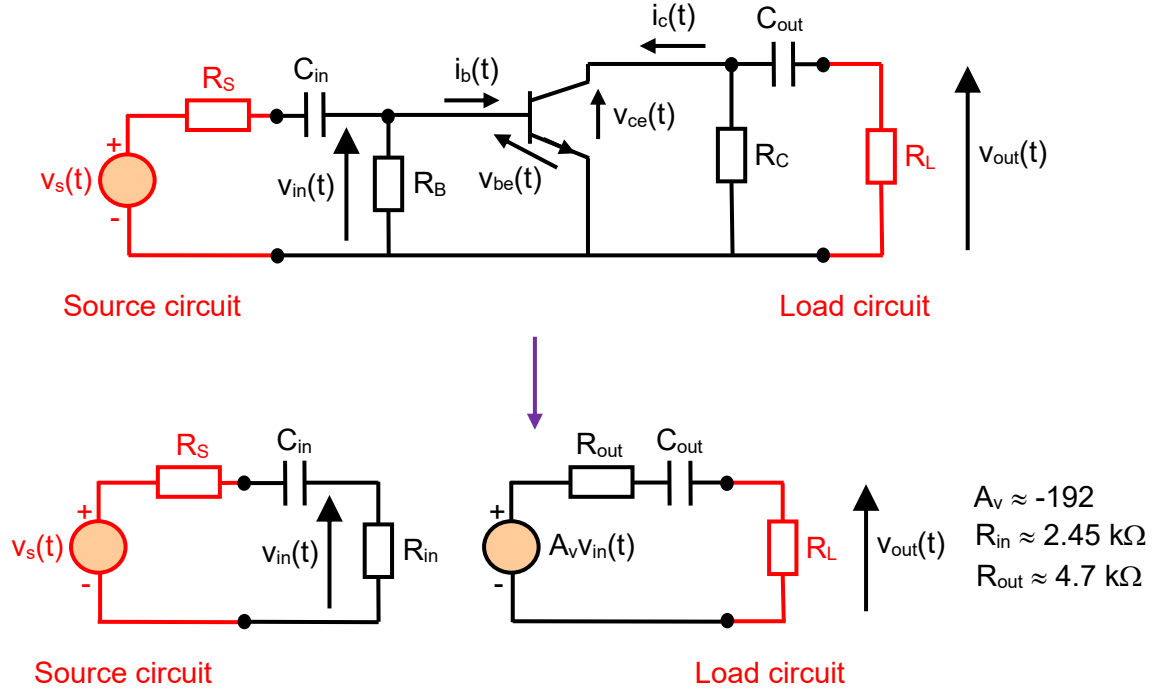
$$\tilde{A}_i \approx 100 \times \frac{5}{2.45 + 5} \times \frac{4.7}{0.01 + 4.7} \approx 100 \times 0.671 \times 1 \approx 67,$$

which indicates that, despite its extremely poor performance in terms of voltage gain, the CE amplifier circuit might still be useful in this situation thanks to its high real current gain.

- **Was it realistic to ignore both coupling capacitors for the AC analysis?**

To answer this question, we must first determine the expression of the real voltage gain $\tilde{A}_v$ obtained when both capacitances C_{in} and C_{out} are taken into account.

We must thus consider the circuit depicted below and assume that $v_s(t)$ is a sinusoidal signal with a frequency ω .



It can be shown that the real voltage gain of the whole circuit, which now depends on the frequency ω of the input signal, is given by

$$\tilde{A}_v(\omega) = \frac{\bar{V}_{out}}{\bar{V}_s} = A_v \frac{R_{in}}{R_{in} + R_s + \frac{1}{jC_{in}\omega}} \frac{R_L}{R_L + R_{out} + \frac{1}{jC_{out}\omega}}$$

$$\Rightarrow \tilde{A}_v(\omega) = A_v \frac{jR_{in}C_{in}\omega}{1 + j(R_{in} + R_s)C_{in}\omega} \frac{jR_L C_{out}\omega}{1 + j(R_L + R_{out})C_{out}\omega}.$$

This equation shows that $\tilde{A}_v(\omega) \rightarrow 0$ as $\omega \rightarrow 0$, which implies that low-frequency signals, and in particular DC signals, cannot be amplified.

For frequencies that are sufficiently high so that $(R_{in} + R_s)C_{in}\omega \gg 1$ and $(R_L + R_{out})C_{out}\omega \gg 1$, the expression of the real voltage gain can be approximated as

$$\tilde{A}_v(\omega) \approx A_v \frac{jR_{in}C_{in}\omega}{j(R_{in} + R_s)C_{in}\omega} \frac{jR_L C_{out}\omega}{j(R_L + R_{out})C_{out}\omega} = A_v \frac{R_{in}}{R_{in} + R_s} \frac{R_L}{R_L + R_{out}},$$

which is the equation previously obtained by ignoring both coupling capacitors.

As a conclusion, it is indeed realistic to ignore the contributions of C_{in} and C_{out} as long as the following conditions are satisfied:

$$\omega \gg \frac{1}{(R_{in} + R_S) C_{in}} \text{ and } \omega \gg \frac{1}{(R_L + R_{out}) C_{out}} .$$

Electronic engineers find it extremely useful to plot the variation of $\tilde{A}_v(\omega)$ as a function of the signal frequency ω . In fact, the complex function $\tilde{A}_v(\omega)$, called *transfer function*, defines the response of the circuit to a given sinusoidal input signal.

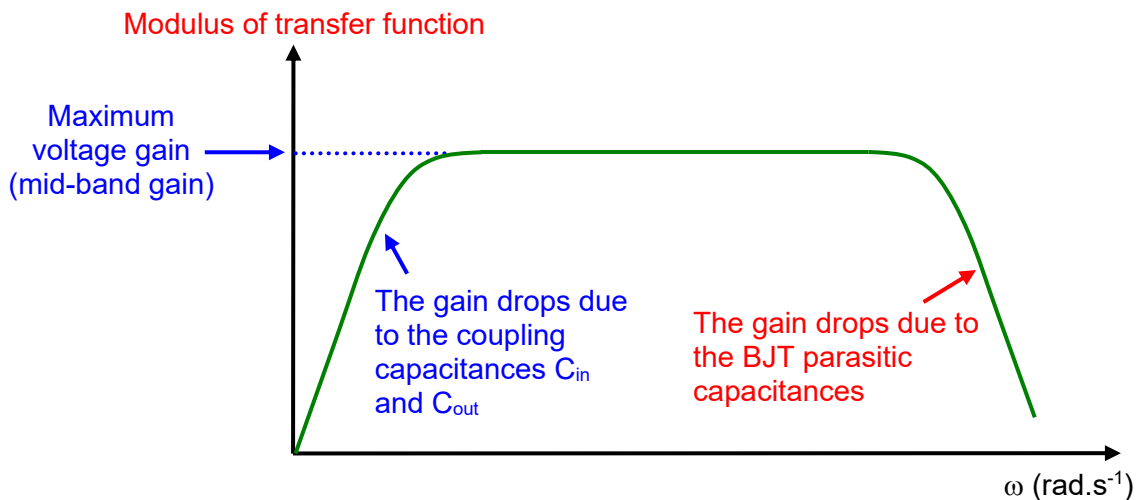
The magnitude of $\tilde{A}_v(\omega)$, i.e. the modulus of the complex function $\tilde{A}_v(\omega)$, shows the effect of the circuit on the amplitudes of sinusoidal input signals at various frequencies.

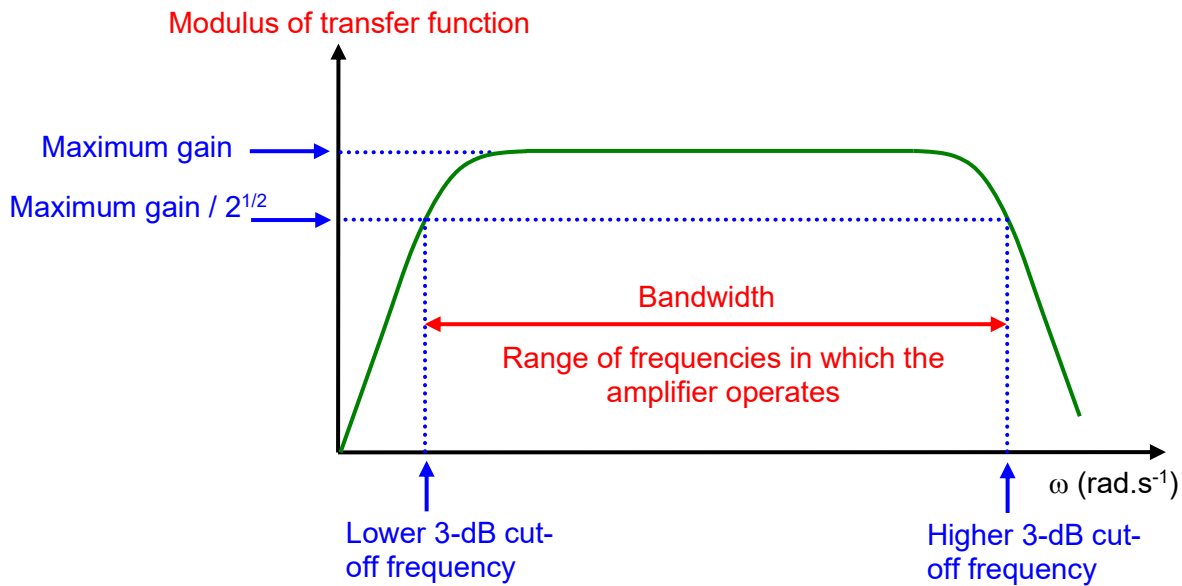
The phase of $\tilde{A}_v(\omega)$, i.e. the argument of the complex function $\tilde{A}_v(\omega)$, indicates the amount of phase shift the circuit introduces on sinusoidal signals as a function of their frequency.

For instance, focusing here on the modulus of $\tilde{A}_v(\omega)$ for our amplifier circuit as an illustration, we obtain

$$|\tilde{A}_v(\omega)| \approx A_v \frac{R_{in} C_{in} \omega}{\sqrt{1 + [(R_{in} + R_S) C_{in} \omega]^2}} \frac{R_L C_{out} \omega}{\sqrt{1 + [(R_L + R_{out}) C_{out} \omega]^2}} .$$

Magnitude of the transfer function for our amplifier (amplitude response)





It is worth mentioning that one can design amplifiers that are able to operate down to DC (0 hertz), i.e. amplify low-frequency signals. In order to do so, one must simply not use any coupling capacitors.

In this case, the biasing circuitry has to be designed by taking into account the source and load circuits, i.e. with the knowledge of their parameters (V_s , R_s , and R_L). This technique for designing amplifier circuits is known as *direct coupling*.

• Device Variability

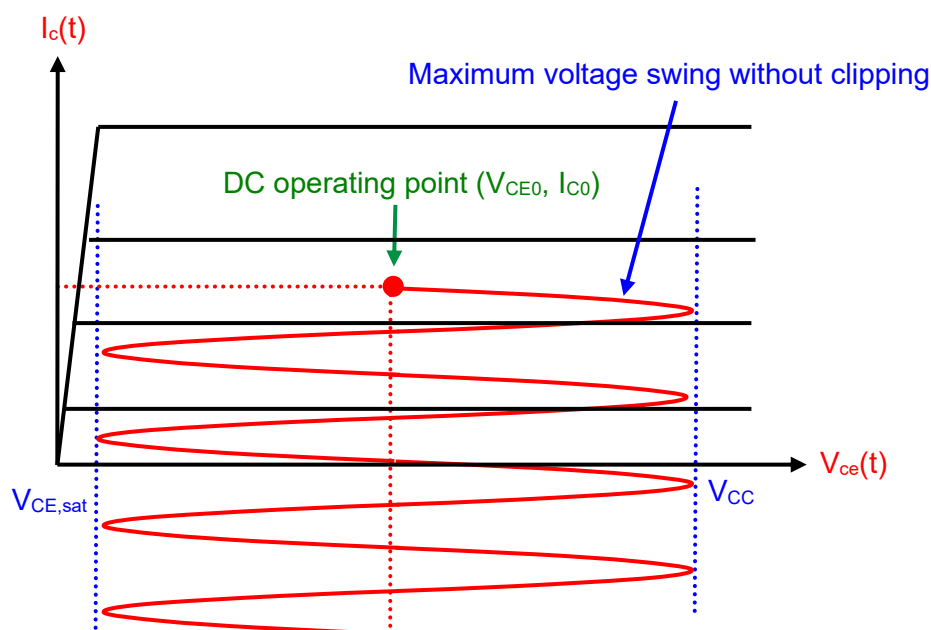
The typical value of the forward current gain β_F for a general-purpose bipolar transistor is in the range 50 to 300. However, this current gain varies considerably with temperature, component aging, and operating conditions.

There is also a considerable spread of characteristics between devices of the same nominal type and even within the same batch. This leads to problems in the design of circuits using BJTs and greatly limits the usefulness of some amplifier circuits, such as the one we have just studied.

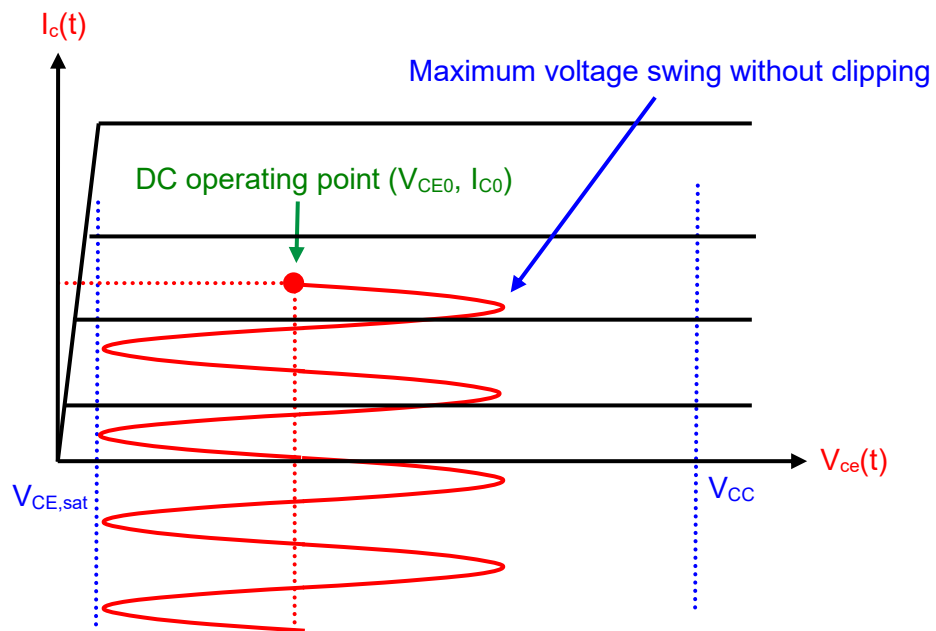
For instance, with our common-emitter amplifier, we have seen that the DC operating points and voltage gain both depend on the value of β_F . The DC analysis of the amplifier was performed assuming $\beta_F = 100$, but this value may not be the exact value of the actual forward current gain of the BJT.

With $\beta_F = 100$, we found that $I_{C0} = \beta_F I_{B0} \approx 1.02 \text{ mA}$ and then $V_{CE0} = V_{CC} - R_C I_{C0} \approx 5.2 \text{ volts}$, which leads to a DC operating point which is positioned midway between $V_{CE,sat}$ and V_{CC} .

This is ideal since it allows the AC signal $v_{ce}(t)$ to swing by nearly $V_{CC}/2 = 5 \text{ volts}$ in either direction.

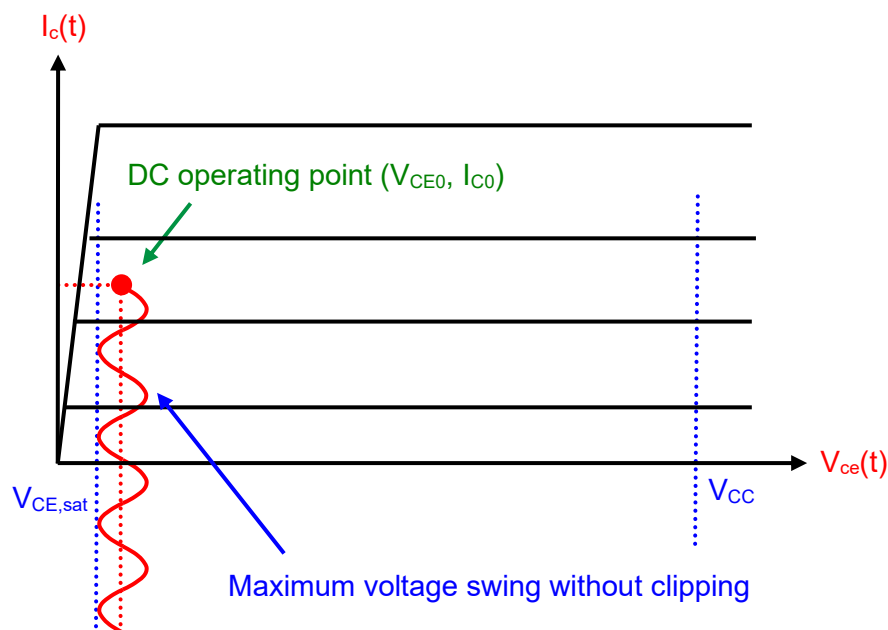


If β_F is actually equal to 150 instead of 100, we can easily see that $I_{C0} = \beta_F I_{B0} \approx 1.53 \text{ mA}$ and then $V_{CE0} = V_{CC} - R_C I_{C0} \approx 2.8 \text{ volts}$, permitting a maximum negative excursion of approximately 2.5 volts before the device enters saturation.



The extreme case is reached with $\beta_F = 200$ instead of 100. Here, the DC output voltage is only $V_{CE0} = V_{CC} - R_C I_{C0} \approx 0.41$ volt which is at the edge of the saturation region, allowing no useful output swing in the negative direction.

For values of the forward current gain above 200, the BJT is driven hard into saturation and the circuit can no longer be used as an amplifier.



One apparent solution to this problem would be to measure the forward current gain of the transistor to be used and design the circuit appropriately. This is however not an attractive solution in a mass-production environment in which it would be impractical to design each circuit uniquely to match individual components. It would also cause severe problems when a device failed and needed to be replaced.

An alternative would be to select devices within a particular close range of parameters. This is possible but expensive, particularly if a very narrow spread is required.

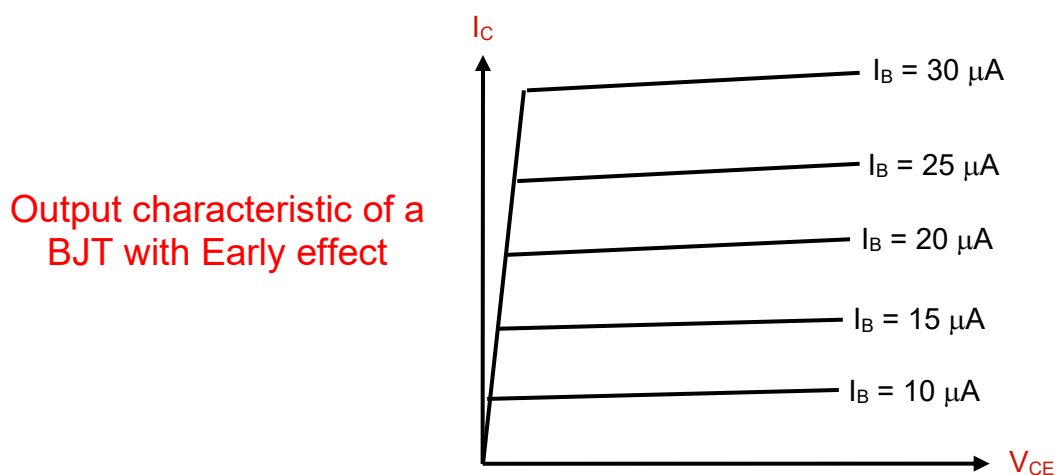
The only practical solution to this problem is to design circuits that are not greatly affected by changes in the forward current gain of the transistors within them. There are techniques to do so, such as negative feedback for instance.

In the tutorials associated with these notes, we will study some amplifier circuits for which the biasing and/or the voltage gain do not depend on the value of β_F .

• Appendix 1: DC Analysis of our CE Amplifier with Early Effect

In the DC analysis of the CE amplifier performed so far, we have ignored the Early effect. We have indeed assumed that the DC collector current I_{C0} only depends on the DC base current I_{B0} , i.e. we have $I_{C0} = \beta_F I_{B0}$.

In some cases, we might want to take into account the Early effect, i.e. the fact that, in the forward active mode, I_{C0} is also slightly dependent on the voltage V_{CE} between collector and emitter.

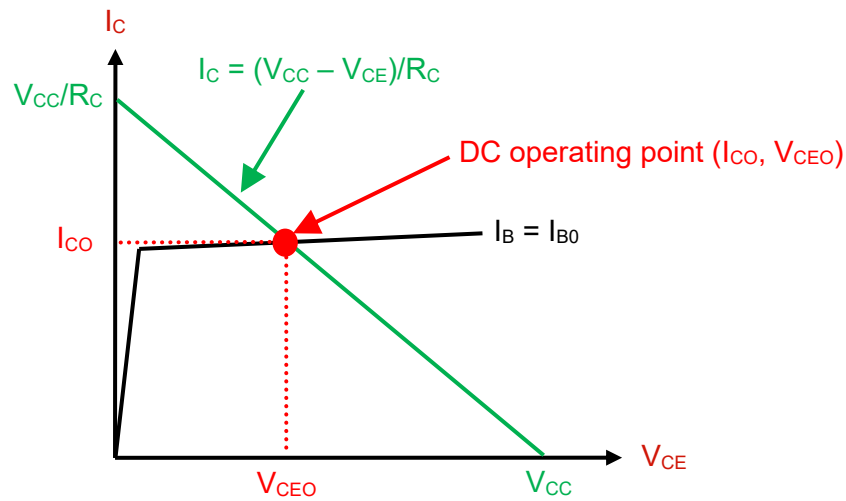


In such case, it is no longer possible to apply the equation $I_{C0} = \beta_F I_{B0}$ in order to compute the value of the collector current.

Instead, we can use the equation $V_{CE0} = V_{CC} - R_C I_{C0}$ which indicates that the relationship between I_{C0} and V_{CE0} is represented by a straight line with the equation

$$I_C = \frac{1}{R_C}(V_{CC} - V_{CE}).$$

The location of the DC operating point is found at the intersection of this straight line with the output characteristic obtained for $I_B = I_{B0}$.



Alternatively, for those who prefer algebra, another technique can be employed: Use two equations to find the location of the DC operating point. These equations are as follows:

(1) $V_{CE0} = V_{CC} - R_C I_{C0}$;

(2) $I_{C0} = \beta_F I_{B0} \left(1 + \frac{V_{CE0}}{V_A} \right)$, where V_A denotes the Early voltage that typically ranges from 50 to 100 volts.

By solving this two-equation system and using $V_A = 75$ volts, $V_{CC} = 10$ volts, $I_{B0} = 10.2 \mu\text{A}$, $R_C = 4.7 \text{ k}\Omega$, and $\beta_F = 100$, we find

$$V_{CE0} = \frac{V_A}{V_A + R_C \beta_F I_{B0}} (V_{CC} - R_C \beta_F I_{B0}) \approx 4.9 \text{ volts}$$

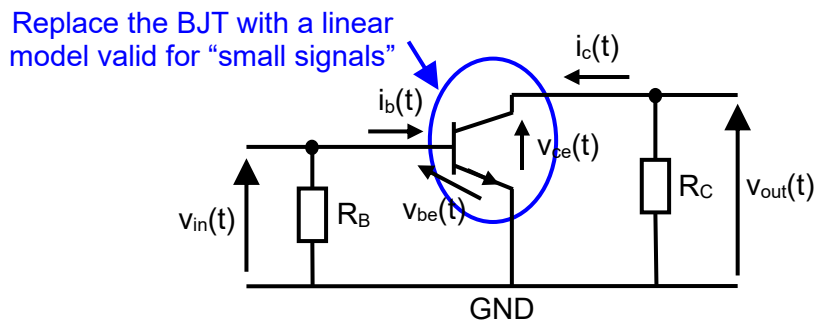
and

$$I_{C0} = \frac{(V_{CC} + V_A)\beta_F I_{B0}}{V_A + R_C \beta_F I_{B0}} \approx 1.09 \text{ mA}.$$

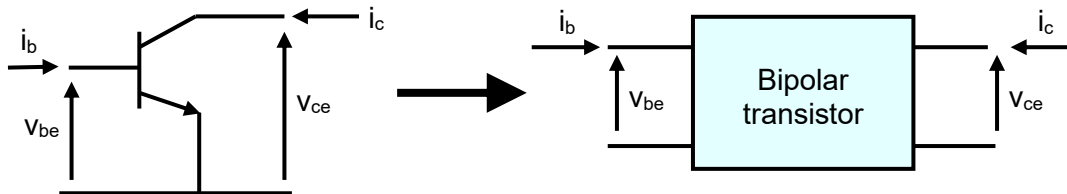
We recall that, by ignoring the Early effect, we found that $V_{CE0} \approx 5.2$ volts and $I_{C0} \approx 1.02$ mA.

• Appendix 2: Complete Small-Signal Model of a BJT

In order to perform the AC analysis of the amplifier, we need to replace the BJT with an equivalent model valid for a small-amplitude input signal (“small signal”). This model is derived assuming that the BJT is in the forward active mode of operation.



The bipolar transistor is now seen as a “black box” with two inputs and two outputs.



The relations between these inputs and outputs are specified using a hybrid matrix (h-matrix):

$$\begin{pmatrix} v_{be} \\ i_c \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \cdot \begin{pmatrix} i_b \\ v_{ce} \end{pmatrix},$$

which is equivalent to $v_{be} = h_{11} i_b + h_{12} v_{ce}$ and $i_c = h_{21} i_b + h_{22} v_{ce}$. The use of an h-matrix is particularly suited to BJTs. The four parameters $h_{i,j}$ are called hybrid parameters (or h-parameters).

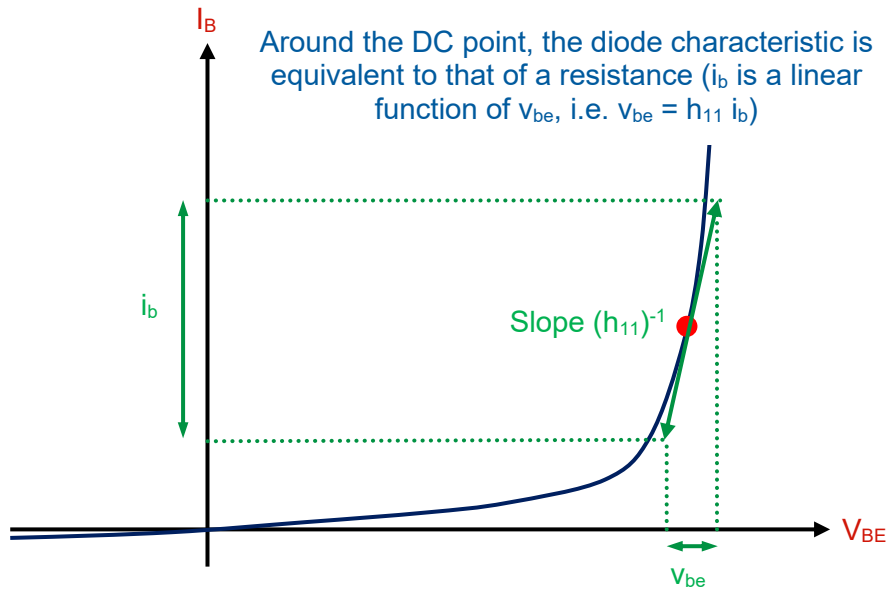
Let us find the expressions of these four parameters for a bipolar transistor.

• **Parameter h_{11}**

$$h_{11} = \left. \frac{v_{be}}{i_b} \right|_{v_{ce} = 0} = \left. \frac{dV_{BE}}{dI_B} \right|_{V_{BE0}, I_{B0}}.$$

Around the DC point, we can consider that the base current i_b varies linearly with the base-emitter voltage v_{be} , i.e. the base-emitter junction is considered as equivalent to a simple resistance h_{11} . The equation of the base-emitter junction characteristic is given by

$$I_B \approx I_S \exp \left\{ \frac{V_{BE}}{V_T} \right\}.$$



Therefore, we can write $\left. \frac{dI_B}{dV_{BE}} \right|_{V_{BE0}, I_{B0}} \approx \frac{I_S}{V_T} \exp \left\{ \frac{V_{BE0}}{V_T} \right\} \approx \frac{I_{B0}}{V_T}.$

At room temperature, the expression of the parameter h_{11} is thus given by

$$h_{11} \approx \frac{V_T}{I_{B0}} \approx \frac{25 \text{ mV}}{I_{B0}}.$$

Since $I_{B0} \approx 10.2 \mu\text{A}$, we have $h_{11} \approx 2.45 \text{ k}\Omega$ when the circuit operates at room temperature.

• **Parameter h_{12}**

$$h_{12} = \left. \frac{v_{be}}{v_{ce}} \right|_{i_b = 0} = 0.$$

In the forward-active mode, the voltage v_{be} only depends on i_b . Thus, if there is no variation of i_b , there is no variation of v_{be} either.

• **Parameter h_{21}**

$$h_{21} = \left. \frac{i_c}{i_b} \right|_{v_{ce} = 0} = \beta_F.$$

The parameter h_{21} is simply the forward current gain of the BJT.

• **Parameter h_{22}**

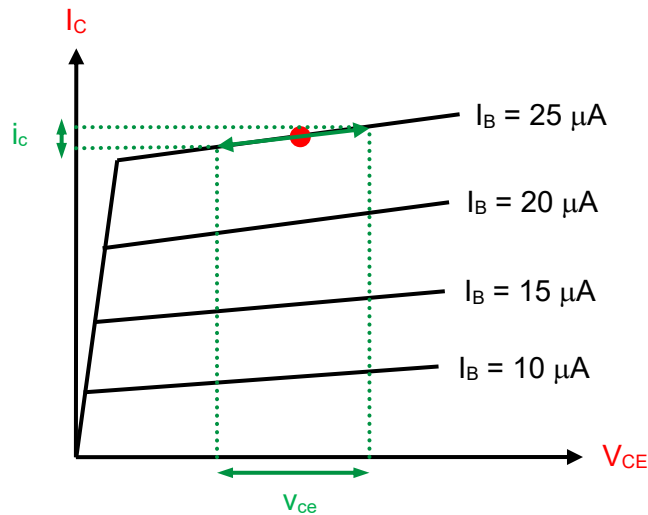
$$h_{22} = \left. \frac{i_c}{v_{ce}} \right|_{i_b = 0}.$$

If we ignore the Early effect, the current i_c only depends on i_b . Thus, if there is no variation on i_b , there is no variation on i_c either. In this case, we obtain

$$h_{22} = \left. \frac{i_c}{v_{ce}} \right|_{i_b = 0} = 0.$$

However, if the Early effect is taken into account, the parameter h_{22} is no longer equal to zero because the collector current i_c is now slightly dependent on the voltage v_{ce} . In such case, h_{22} is the slope of the “ I_C versus V_{CE} ” characteristic around the DC operating point:

$$h_{22} = \left. \frac{i_c}{v_{ce}} \right|_{i_b = 0} = \left. \frac{dI_C}{dV_{CE}} \right|_{V_{CE0}, I_{C0}}.$$



To determine h_{22} , we can use the expression of the collector current that takes the Early effect into account:

$$I_C = \beta_F I_B \left(1 + \frac{V_{CE}}{V_A} \right),$$

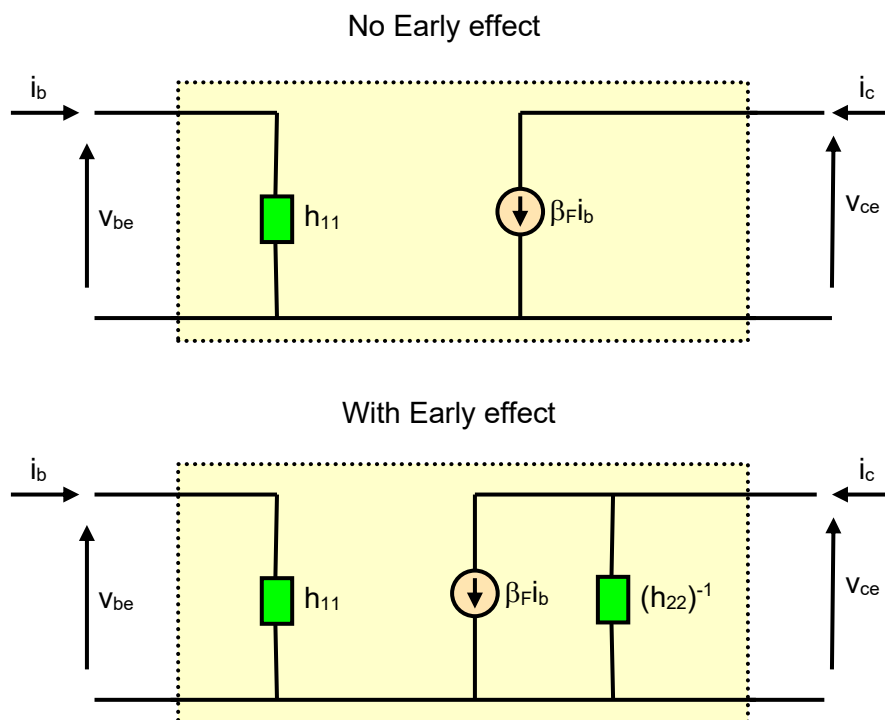
where V_A denotes the Early voltage that typically ranges from 50 to 100 volts.

Thus, the expression of h_{22} is given by $\left. \frac{dI_C}{dV_{CE}} \right|_{V_{CE0}, I_{C0}} = \frac{\beta_F I_{B0}}{V_A}$.

For $I_{B0} \approx 10.2 \mu\text{A}$, $\beta_F = 100$ and V_A ranging from 50 to 100 V, the value of h_{22} ranges from $10.2 \mu\text{S}$ to $20.4 \mu\text{S}$. This corresponds to a resistance $(h_{22})^{-1}$ ranging from approximately 50 to 100 $\text{k}\Omega$.

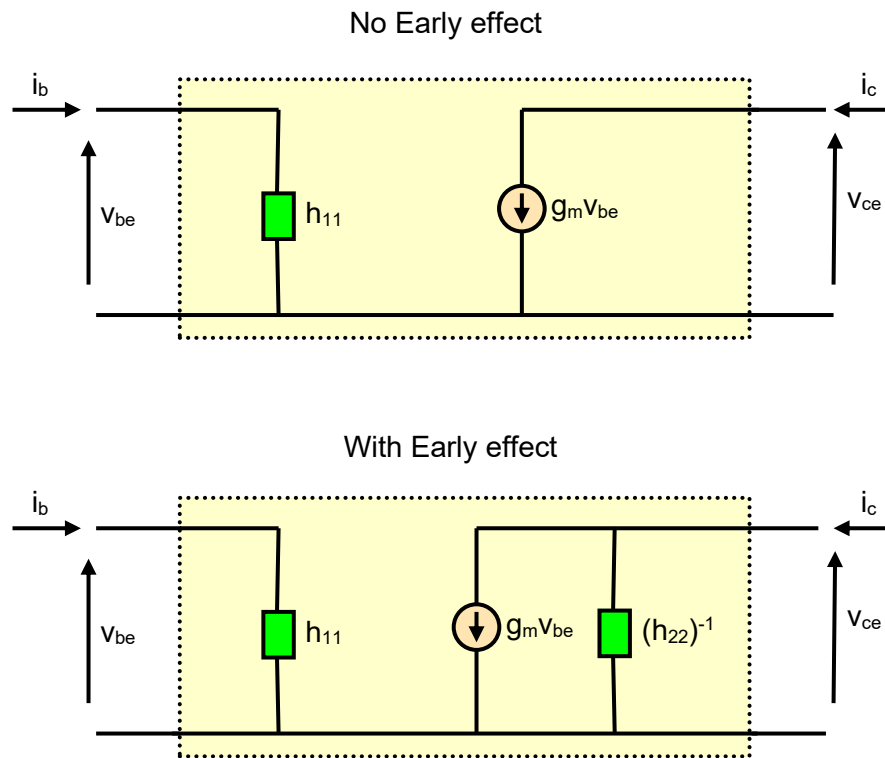
In many cases, the resistance $(h_{22})^{-1}$ can be ignored due to its very high value, as if there was no Early effect at all.

We finally obtain the small-signal model of a BJT shown below.



This is the model of a current amplifier for which h_{11} is the input resistance, $(h_{22})^{-1}$ is the output resistance, and β_F is the unloaded current gain. In the model without Early effect, we simply assume that $(h_{22})^{-1} \rightarrow +\infty$ ($h_{22} = 0$).

In some textbooks, you will often find an alternative small-signal model for a BJT. This model is as shown below.



In this alternative model, the collector current i_c is seen as a function of v_{be} rather than i_b . The parameter g_m is called transconductance and is expressed in siemens (S). It is computed using

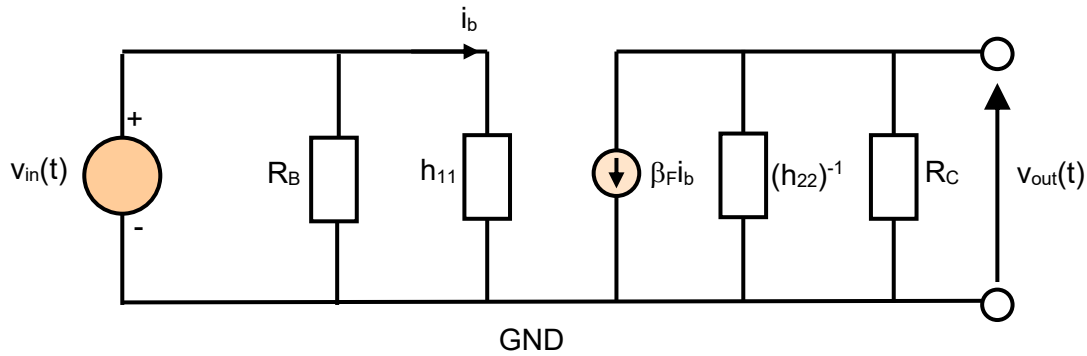
$$g_m = \left. \frac{i_c}{v_{be}} \right|_{v_{ce} = 0} = \left. \frac{i_c}{i_b} \frac{i_b}{v_{be}} \right|_{v_{ce} = 0} = \frac{\beta_F}{h_{11}} \approx \beta_F \frac{I_{B0}}{V_T} = \frac{I_{C0}}{V_T}.$$

At room temperature, typical values of the transconductance g_m range from 40 mS (for $I_{C0} = 1$ mA) to 400 mS (for $I_{C0} = 10$ mA). This small-signal model is popular with many electronic engineers because it is very similar to that of a field-effect transistor and therefore allows for a direct comparison between both types of transistors based on their respective transconductances.

Note that both small-signal BJT models are obviously strictly equivalent. It means that the choice of either of them when performing an AC analysis will not affect the final results.

• Appendix 3: AC Analysis of our CE Amplifier with Early Effect

We can replace the bipolar transistor with its small-signal model in order to determine the expressions of the voltage gain, input resistance, and output resistance of our CE amplifier.



We recall the values of the various parameters: $R_B = 910 \text{ k}\Omega$, $h_{11} \approx 2.45 \text{ k}\Omega$, $\beta_F = 100$, $R_C = 4.7 \text{ k}\Omega$. For this calculation, we take the Early effect into account and assume $(h_{22})^{-1} = 75 \text{ k}\Omega$.

• Unloaded voltage gain

We can show that $v_{out}(t) = -\frac{R_C}{1 + h_{22} R_C} \beta_F i_b$. Therefore, we obtain

$$v_{out}(t) = -\frac{R_C}{1 + h_{22} R_C} \beta_F \frac{v_{in}(t)}{h_{11}}.$$

The unloaded voltage gain of our CE amplifier is finally given by

$$A_v = \frac{v_{out}(t)}{v_{in}(t)} = -\beta_F \frac{R_C}{h_{11}(1 + h_{22} R_C)} \approx -180.$$

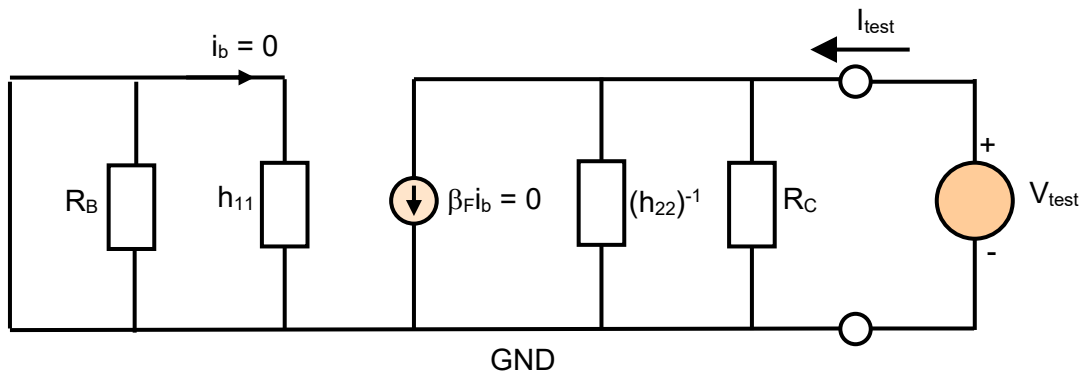
If we ignored the Early effect ($h_{22} = 0$), we would obtain $A_v = -\beta_F \frac{R_C}{h_{11}} \approx -192$.

• Input resistance

The input resistance is given by $R_{in} = \frac{R_B h_{11}}{R_B + h_{11}} \approx h_{11}$ since $h_{11} \ll R_B$. Thus, we have $R_{in} \approx 2.45 \text{ k}\Omega$.

• Output resistance

The output resistance is given by $R_{\text{out}} = \frac{R_C}{1 + h_{22} R_C} \approx 4.4 \text{ k}\Omega$.



If we ignored the Early effect, we would obtain $R_{\text{out}} = R_C = 4.7 \text{ k}\Omega$.